Nama: Nindira adiarty

NIM: 2510514006

Mata Kuliah: Matematika Sains Data

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Parameter Personalisasi

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NIM: 2510514006 (Nindira adiarty)

Simbol	Definisi	Rumus	Nilai
N	Dua digit akhir	2510514006 mod 100	N = 06
a	Digit puluhan NIM	floor (06/10)	a = 0
b	Digit satuan NIM	06 mod 10	b = 6
K	Konstanta skala	(a + b + 1) = (0 + 6 + 1)	K = 7
θ_0 (dimapat)	Sudut awal bil. kompleks	(a · 30) + (b · 5)	$\theta_0 = 30°$
α	Learning rate gradient	0,001 × (b + 1)	α = 0,007

Soal 3 Nindira adiarty 2510514006

Definisi himpunan pelanggan

$A = \{N, N+2, N+4, N+7, N+10, N+13, N+15\}$ dari medsos

$B = \{N+3, N+5, N+7, N+10, N+12, N+18\}$ dari referral

$C = \{N+1, N+7, N+10, N+11, N+17, N+20\}$ dari loyalty program

Universe $U = \{N, N+1, N+2 \ldots N+20\}$ (total 21 pelanggan)

* Subs $N = 6$, maka:

$A = \{6, 8, 10, 13, 16, 19, 21\}$ ← medsos

$B = \{9, 11, 13, 16, 18, 24\}$ ← referral

$C = \{7, 13, 16, 17, 23, 26\}$ ← loyalty program

$U = \{6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26\}$

* Operasi Himpunan

- $A \cap B = \{13, 16\}$ - $A \cap C = \{13, 16\}$ - $B \cap C = \{13, 16\}$ - $A \cap B \cap C = \{13, 16\}$

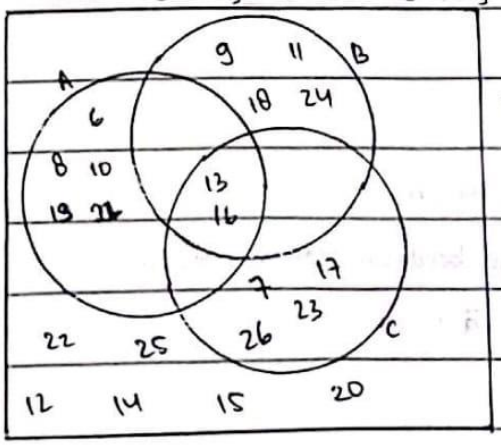

Diagram Venn tiga himpunan (a)

$|A| = \{6, 8, 10, 13, 16, 19, 21\} = 7$ $|A \cap B| = |A \cap C| = |A \cap B| = 2$

$|B| = \{9, 11, 13, 16, 18, 24\} = 6$ $|A \cap B \cap C| = 2$

$|C| = \{7, 13, 16, 17, 23, 26\} = 6$

$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

$= 7 + 6 + 6 - 2 - 2 - 2 + 2 = \underline{\underline{15}}$

$|A \cap B \cap C| = \{13, 16\} = \underline{\underline{2}}$

$|(A \cup B) \cap C'| =$

$(A \cup B) = \{6, 8, 9, 10, 11, 13, 16, 18, 19, 21, 24\}$

$C' = \{6, 8, 9, 10, 11, 12, 14, 15, 18, 19, 20, 21, 22, 24, 25\}$

maka $|(A \cup B) \cap C'| = \{6, 8, 9, 10, 11, 18, 19, 21, 24\} = \underline{\underline{9}}$

$|A \oplus B|$ (Simetris difference = $(A \cup B) - (A \cap B)$)

$(A \cup B) = \{6, 8, 9, 10, 11, 13, 16, 18, 19, 21, 24\}$

$(A \cap B) = \{13, 16\}$

maka $|A \oplus B| = \{6, 8, 9, 10, 11, 18, 19, 21, 24\} = \underline{\underline{9}}$

c) diberikan 2 fungsi

$f(u) = K \cdot u + (a + b)$ → harga eceran

$g(u) = \dfrac{u^2}{(b+1) + a}$ → Surcharge berdasar Volume pembelian

Subs $K = 7$ $b = 6$ $a = 0$

$f(u) = 7u + 6$ $g(u) = \dfrac{u^2}{7}$

Tentukan

1) $(F \circ g)(u) = f(g(u)) = f\left(\frac{u^2}{7}\right) = \cancel{7} \cdot \frac{u^2}{\cancel{7}} + 6 = u^2 + 6$

$(g \circ f)(u) = g(f(u)) = g(7u+6) = \frac{(7u+6)^2}{7} = \frac{49u^2 + 84u + 36}{7}$

$= 7u^2 + 12u + 36/7$

2) Apakah komposisi komutatif? buktikan dgn cth numerik. $u = 6+2$

- $(F \circ g)(6+2) = \cancel{\ldots} = 8^2 + 6 = 64 + 6 = 70$
- $(g \circ F)(8) = 7(8)^2 + 12(8) + \frac{36}{7} = 448 + 96 + 5,1429 = 549,1429$

karena $70 \neq 549,1429$

maka $(F \circ g) \neq (g \circ F)$ dan komposisi tidak komutatif.

d) Tentukan $f^{-1}(u)$

$y = f(u) = 7u + 6 \rightarrow u = Fy + 6$

$u - 6 = 7y$

$y = \frac{u-6}{7} = \boxed{F^{-1}(u) = \frac{u-6}{7}}$

Verif = $f(F^{-1}(u)) = \frac{\cancel{7}(u-6)}{\cancel{7}} + 6 = u$

Pendapatan 1 pelanggan Rp $(K \cdot 100 + a + b)$, berapa volume pembelian u yang menghasilkannya?

Rp. $((k \cdot 100) + a + b)$ ribu

Rp. $((7 \cdot 100) + 0 + 6) =$ Rp. 706 ribu

$$u = f^{-1}(706) = \frac{706 - 6}{7} = \frac{700}{7} = 100$$

Maka Volume pembelian yang menghasilkan Rp. 706 ribu adalah $u = 100$ unit .

Verif: $f(100) = 7(100) + 6 = 706 \rightarrow$ betul.

e) klasifikasi fungsi f dan g

$f(u) = 7u + 6$

mis $u = i \rightarrow 7 \cdot i_1 + 6 = 7 \cdot i_2 + 6$

$7 \cdot i_1 = 7 i_2$

$i_1 = i_2 \Rightarrow$ karena selalu mengimplikasi

Maka f adalah injektif

ambil sembarang $y \in R$. Apakah selalu $u \in R$ hingga $f(u) = y$?

$7u + 6 = y \Rightarrow u = \frac{y-6}{7} \in R \Rightarrow$ Maka f adalah surjektif.

karena f mengandung keduanya,

f adalah fungsi Bijetif (Injektif dan Surjektif)

fungsi $g(x) = \frac{x^2}{7}$

misal $x_1 = 3$ $x_2 = -3$ keduanya $\in R$

$g(3) = \frac{9}{7}$ $g(-3) = \frac{3}{7}$ $g(3) = g(-3)$

tapi $3 \neq 3$, maka g bukan injektif.

misal $y = -1 \in R$. Apakah $x \in R$ hingga $g(x) = -1$

$\frac{x^2}{7} = -1$ $x^2 = -7$ karena tdk ada solusi untuk $x^2 = -7$

maka $y = -1$ tdk ada preimage di R

bukan surjektif.

Karena g tidak mengandung keduanya,

g bukan fungsi injektif, surjektif dan bijektif.